

# Henry Zhang

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## EXPERIENCE

### U.S. Department of the Treasury

June 2026 – Present

*Software Engineer*

*Washington, DC*

- Architected an event-driven mailbox triage system on AWS with webhook ingestion, failure reconciliation, AI classification, and information extraction, saving ~10 staff-hours per week while automating leadership reporting.
- Unified ~20K financial records weekly across disparate sources using entity deduplication and a shared ontology, and built graph-based relationship models to surface suspicious transactions and hidden entity connections.
- Built an AI security evaluation platform that ran reproducible model evaluations against standardized benchmarks in isolated Docker sandboxes and parallel GCP VMs to assess model exploitability.
- Built and productionized a Microsoft Graph-backed MCP integration for Treasury's internal ChatGPT ecosystem, implementing per-user authentication, data isolation, and request throttling under a 4 req/s service constraint for 4,000+ staff.

### Trianz Inc.

September 2025 – June 2026

*Associate Project Manager*

*Herndon, VA*

- Managed migration readiness for 60+ applications moving to AWS, translating technical dependencies and deployment blockers into migration plans across Engineering, DevOps, security, and client stakeholders.
- Identified manual reporting bottlenecks and built a full-stack AI application that automated metrics from Excel project trackers and used LLMs to extract risks, blockers, and status updates from unstructured meeting notes, enabling on-demand reporting.

### Medidata Solutions Inc.

June 2024 – August 2024

*AI Software Engineering Intern*

*New York, NY*

- Built a retrieval-augmented data pipeline combining embedding-based similarity search with keyword filtering to enable high-recall retrieval across 40,000+ medical records.
- Established an evaluation framework for the RAG system using precision, recall, and AI-assisted relevance scoring to benchmark retrieval accuracy, coverage, and trustworthiness
- Optimized PostgreSQL hybrid search by filtering candidates with full-text search before PGvector similarity ranking and reducing unnecessary joins and intermediate data transfer.

### Microsoft AI Co-Innovation Labs

June 2023 – August 2023

*AI Software Engineering Intern*

*Shanghai, China*

- Developed an enterprise LLM chatbot using embeddings, prompt engineering, and agent workflows to enable natural-language querying across internal company datasets, cutting PM reporting cycles from weeks to days.
- Accelerated go/no-go decisions for client AI and Azure initiatives by translating business needs into scoped Azure and machine-learning pilots with defined data requirements, feasibility criteria, and implementation tradeoffs.

## PROJECTS

### Inyo | Founding Engineer & Technical Lead, Part-Time

April 2026 – June 2026

- Architected and launched an AI-powered SMS matchmaking platform supporting thousands of users, building the event-driven backend, AI conversation and profile-extraction pipelines, admin tooling, and production infrastructure.
- Led three engineers and partnered with the founder to scope MVP priorities, translate product requirements into technical workstreams, and rapidly deliver features while balancing scalability, reliability, and execution speed.

### Fine-tuning and Supervised Learning for Stock Prediction | Machine learning, Data

2024

- Increased labeling accuracy on financial news by 9% by fine-tuning LLMs with qLoRA, improving accuracy from 0.75 to 0.84 to provide more accurate insights in market trend prediction

## EDUCATION

### New York University

New York, NY

*BS in Computer Science, Minor in Business*

## TECHNICAL SKILLS

**Languages:** Python, Java, SQL, JavaScript, HTML/CSS, C/C++

**Tools/Libraries/Frameworks:** Git, Docker, React, AWS/Azure/GCP, Excel, Postgres, MongoDB, Redis, BI Tools, Pandas/NumPy, Langchain, pyTorch, MERN Stack, Node.js, Flask, Django, **AWS Cloud Practitioner**.

**Skills:** Documentation, Project Management, Cross-functional Collaboration, Client Communication, Stakeholder Management